Last Meeting (Feb 27)

What we discussed:

- Presented differentiated-products model (logit demand) with switching costs and asymmetric information
- Identification strategy: 4-step sequential procedure using $(r_i^{\text{win}}, W_i, D_i)$
- To do: add consumer heterogeneity

Today's meeting:

- New information structure: bank 1 observes default probability h, bank 2 observes noisy signal $\mathbf{x}$
- Full identification — nonparametric, no parametric assumptions on $F(h|\mathbf{x})$ or G

Model Setup

Two banks compete for a borrower

	Bank 1 (incumbent)	**Bank 2 (entrant)**
Default probability h	Observed	Not observed
Covariates $\mathbf{x} \in \{x_1, \ldots, x_K\}$	**Not observed**	Observed

- Logit demand: $s(r_1, r_2) = \frac{1}{1+\exp(\alpha(r_1 - r_2) - \lambda)}$

Key consequence: Bank 1 cannot condition on $\mathbf{x}$, so $\sigma^*(h)$ depends on h alone

$\Rightarrow$ **One-to-one** mapping from bank-1 rates to borrower types

Equilibrium

Bank 1 observes h, forms posterior $\pi_j(h) = \Pr(\mathbf{x} = x_j \mid h)$ via Bayes' rule, and maximizes expected profit over the K competitive environments:

$$\boxed{\sigma^*(h) = \frac{1}{1-h} + \frac{1}{\alpha} \cdot \frac{\bar{S}(\sigma^*(h);\, h)}{\bar{V}(\sigma^*(h);\, h)}}$$

- $\bar{S} = \sum_j \pi_j(h)\, s(\sigma^*(h), r^*_{2,j})$: expected market share (averaged over competitive environments)
- $\bar{V} = \sum_j \pi_j(h)\, s(\sigma^*(h), r^*_{2,j})(1 - s(\sigma^*(h), r^*_{2,j}))$: expected share variance

Data: For each borrower i, the econometrician observes:

- r_i^{win}: winning interest rate
- $W_i \in \{1, 2\}$: identity of the winning bank
- $D_i \in \{0, 1\}$: default indicator

Identification: Type Recovery and Nonparametric Objects

Since $\sigma^*(h)$ is one-to-one, among bank-1 winners:

$$E[D_i \mid r_i^{\text{win}} = r_1,\ W_i = 1] = \tau(r_1) = h$$

Object	**Identified from**
$\sigma^*(h)$, $\tau(r_1)$	$E[D \mid r_1, W = 1]$ (type recovery)
$m(h) = \sigma^*(h) - \frac{1}{1-h}$	Directly from $\sigma^*(h)$
$r_{2,j}^*$ (distinct bank-2 rates)	Bank-2 winning rates take K distinct values
$E[D \mid r_2, W = 2]$	Regression among bank-2 winners

What remains: (α, λ), population probabilities (p_j), type distributions $f(h|x_j)$

Identification: Two Restrictions at Each h

Define $\gamma_j(h) \equiv p_j\, f(h|x_j)$: the joint density of (x_j, h) and $s_j(h) \equiv s(\sigma^*(h), r^*_{2,j};\, \alpha, \lambda)$. At each h, two restrictions:

(R1) First-order condition:

$$\sum_{j=1}^{K} \gamma_j(h)\, s_j(h) \Big[1 - \alpha\, m(h)\, \big(1 - s_j(h)\big)\Big] = 0$$

(R2) Law of total probability for bank-1 winners:

$$\sum_{j=1}^{K} \gamma_j(h)\, s_j(h) = f(h,\ W_i = 1)$$

Right-hand side is data: type recovery gives $h_i = \tau(r_i^{\text{win}})$ for each bank-1 winner, so the empirical distribution of h among stayers directly estimates the joint density $f(h,\ W_i = 1)$.

$\Rightarrow$ **2 equations in K unknowns** $\gamma_1(h), \ldots, \gamma_K(h)$ at each h

- $K = 2$: exactly identified pointwise $\Rightarrow$ **nonparametric** identification

Case $K \geq 3$: Parametric Identification

2 equations, K unknowns per $h \Rightarrow (K-2)$ free functions $\Rightarrow$ nonparametric ID fails.

Resolution: parametrize $f(h|x_j) = f(h;\, \theta_j)$ with a P-parameter family (e.g., Beta: $P = 2$).

(R2) becomes: $\sum_{j=1}^{K} p_j\, f(h;\theta_j)\, s_j(h) = f(h,\, W_i{=}1)$ for all h

Identification (same sequential logic as $K = 2$):

1. Given (α, λ): continuum of equations pins down $\underbrace{KP}_{\substack{P \text{ params per} \\ \text{density} \times K \text{ groups}}} + \underbrace{K-1}_{\substack{p_1,\ldots,p_{K-1} \\ (p_K = 1 - \sum p_j)}}$ scalar unknowns
2. (α, λ) pinned down by the K bank-2 default-rate conditions ($K \geq 3 > 2$ unknowns)

Reduced-Form Evidence: REDEC Policy (Chile, April 2026)

Setting: Chile creates a centralized debt registry (REDEC) — all lenders must share positive & negative credit info. Non-bank borrowers' histories become visible to banks for the first time.

Regressions (sample: non-bank-only borrowers, $Good_i = \mathbf{1}[\text{no delinquencies}]$):

1. Credit access:

$$NewBank_{it} = \beta^A \cdot (Post_t \times Good_i) + \gamma X_{it} + \delta_i + \theta_t + \varepsilon_{it}$$

Expected: $\hat{\beta}^A > 0$ — good types gain bank access once histories are visible

2. Pricing:

$$r_{ijt} = \beta^P \cdot (Post_t \times Good_i) + \gamma X_{it} + \mu_j + \theta_t + \varepsilon_{ijt}$$

Expected: $\hat{\beta}^P < 0$ — good types get lower rates (separating vs. pooling)

3. Delinquency:

$$Delinq_{ijt} = \beta^D \cdot (Post_t \times OCNB_Debt_i) + \gamma X_{it} + \mu_j + \theta_t + \varepsilon_{ijt}$$

Expected: $\hat{\beta}^D < 0$ — banks screen on non-bank debt, improving loan quality

Appendix: Case $K = 2$ — Nonparametric Identification

Step 1: Given candidate (α, λ), solve (R1)–(R2) — a 2×2 system at each h.

Since $r^*_{2,1} \neq r^*_{2,2} \Rightarrow s_1(h) \neq s_2(h) \Rightarrow$ system is nonsingular $\Rightarrow \gamma_j(h;\, \alpha, \lambda)$ determined at every h.

Step 2: Recover p_j: integrability $\int f(h|x_j)\, dh = 1$ gives $p_j = \int \gamma_j(h)\, dh$.

Step 3: Pin down (α, λ) using bank-2 conditional default rates:

$$E[D_i \mid r^*_{2,j},\ W_i = 2] = \frac{\int h\,(1 - s_j)\, f(h|x_j)\, dh}{\int (1 - s_j)\, f(h|x_j)\, dh}, \quad j = 1, 2$$

Two equations, two unknowns $\Rightarrow (\alpha, \lambda)$ identified.